BREAST CARE-MA DIAGNOSTIC DECISION SUPPORT REPORT

Department of Information Technology | Nehru Institute of Engineering and Technology
Explainable Multi-Agent AI System for Ultrasound Modality

Case ID:	BUSI-BENIGN-5CE5	**Date & Time:**	2026-10-31 00:13:22
Patient Age:	48	**Family History:**	No
Breast Density:	C - Heterogeneously dense	**Symptoms:**	Palpable nodule

1. Quantitative AI Findings & Model Outputs

Parameter	Result	Clinical Meaning
AI Classification	**NORMAL**	Calibrated Malignant Probability: 8.2%
Tumor Detection	Lesion Detected	Relative Area: 37.10%
Morphology (Agent 3)	Irregular / Spiculated	Aspect Ratio: 1.32, Solidity: 0.63
Acoustic Shadowing	Absent	Acoustic attenuation distal to mass
ACR BI-RADS Category	**BI-RADS 1**	**Negative / Routine Screening**

2. Retrieved Evidence & Clinical Guidelines (Agent 4)

• **Normal Ultrasound Examination** (ACR BI-RADS Atlas 5th Edition (Ultrasound)): No masses, architectural distortion, or suspicious calcifications are detected. Routine age-appropriate screening is recommended.

• **Classic Malignant Presentation (>=95% Probability)** (ACR BI-RADS Atlas 5th Edition / Society of Breast Imaging): Classic presentation: Spiculated margins, taller-than-wide (non-parallel) orientation, hypoechoic with acoustic shadowing, microcalcifications. Malignancy risk exceeds 95%.

3. Recommended Clinical Follow-up Actions

• Routine annual screening mammography/ultrasound as per age guidelines.

• Clinical breast examination (CBE) during annual wellness visit.

REGULATORY & SAFETY DISCLAIMER: NOTICE: Breast CARE-MA is an AI decision-support research prototype developed for academic evaluation. It does not provide definitive medical diagnosis and must never supersede clinical judgment by a licensed radiologist or oncologist.